

Folio Test Plan

Nicholas Dunzelman's Contribution

Workflows, exports, and privacy

Use a separate test database in the local Streamlit application. Create plan A with two tasks and plan B with one task. Use distinct task names, known times, and saved completion states. Open exported PDF and DOCX files in their viewers and inspect the prepared email text. Each case passes only when all stated expected results occur. These are planned acceptance tests, not execution results.

N1 Plan creation and editing (F1–F3, P1)

Fixture & Approach: Select an assignment, a test, and another coursework item in separate runs. Choose a template, edit a suggested task, and save. Reopen each plan.

Expected: the chosen coursework, template, and edited task are preserved. Change the edited task again and save a revision; reopening shows the latest saved values, with the other plans unchanged.

N2 Instructor suggestions (F4–F7, P1)

Fixture & Approach: On separate copies of plan A, enter the suggestion “Review chapter 2.” Accept and save it; reject it; adapt it to “Review chapter 3” and save; then accept but decline to save a revision.

Expected, respectively: the original suggestion is applied, the plan is unchanged, only the adapted task is applied, and the saved plan is unchanged. Reopen each saved plan to verify.

N3 Daily, weekly, and rescheduled tasks (F8–F10)

Fixture & Approach: Schedule A1 Monday at 10:00–11:00, A2 Tuesday at 10:00–11:00, and B1 Monday at 11:00–12:00.

Expected: Monday’s daily view shows A1 and B1; the weekly view includes all three. Move A1 to 09:00–10:00. Both views show the new time and retain task identity. Attempt to move A1 to 11:30–12:30. B1 is identified as conflicting, and the saved time remains 09:00–10:00.

N4 Reminders (F11)

Fixture & Approach: Enable a reminder for A1 at an accepted time and leave Folio running through that time.

Expected: the configured reminder appears. Change the time, verify the saved setting, disable the reminder, and verify that no reminder appears at the previously configured time. Submit 25:00 as the reminder time: a validation error appears, and the saved setting remains unchanged.

N5 Completion and progress (F12–F14, P3)

Fixture & Approach: With both A tasks incomplete, complete A1, repeat the completion action, complete A2, then mark A1 incomplete.

Expected: A’s progress bar shows 0%, 50%, 50%, 100%, then 50%; congratulations appear on each change to complete. B’s progress bar remains unchanged. Reopen A and verify saved task states match the display. An empty plan displays zero progress without an error.

N6 Independent planning (F15–F16, F48–F52)

Fixture & Approach: With no instructor feedback, create, view, edit, save, reschedule, and complete a plan without sharing it. Repeat with game and rewards disabled.

Expected: all planning actions remain available, and saved coursework, plans, and tasks remain intact when those optional features are disabled.

N7 Limits on inferred activity (F17)

Fixture & Approach: Enter a low score and an overdue coursework item, then inspect the plan and schedule.

Expected: Folio presents them as planning context and makes no claim about studying, attendance, participation, or effort outside Folio, and marking a task complete changes only the recorded in-app task status and its associated progress/rewards.

N8 Selected PDF and DOCX content (I1–I2, I4, P2)

Fixture & Approach: Select the course, the coursework name, and one planned task to share; leave the other task unselected. Export PDF and DOCX.

Expected: both files open and contain exactly the selected academic values, including edited text, with the unselected task absent. Repeat after deselecting all academic fields: neither output may include previously selected or private values.

N9 Export privacy (I5, F47)

Fixture & Approach: Give progress, XP, points, level, badge, cosmetic, and game fields distinct private test values. Run PDF, DOCX, and email preparation, including a selection that attempts to include a private field. Inspect visible output, extracted text, document properties, DOCX package contents, and email subject/body/attachments.

Expected: every private field and test value is absent; selected academic content remains. Repeat after changing private values; academic output content is unchanged.

N10 Email review (I3–I5, P2)

Fixture & Approach: Prepare an email from the N8 selection using a test recipient.

Expected: an unsent preview contains exactly the selected academic values and no private data. Edit the preview, then cancel. Verify no message was sent. Prepare it again and inspect the revised preview without sending; preparation alone must never send a message.

Caleb Brooks' Contribution

Data, Scheduling, and Game Correctness

Use a detached test database in the local Streamlit application. Reset the DB to the stated fixture (prepared data used to run a test) before each independent test. Use a course with plan A and tasks A1 and A2, and another course with plan B and task B1. Record task identifiers and their saved values for later checking changes against the DB and the interface. Enter dates and construct CSV files using the Folio interface-documented formats.

Run workflow tests through the application. Use an automated test suite for validation, overlap computation, and repeated reward events (where available). Record the results, supporting screenshots, DB checks, and pass-or-fail status for each case.

C1 Manual Coursework Creation, Editing, and Deletion (F18–20, F53, I12–13)

Fixture & Approach: Create coursework with a course name, assignment name, and valid due date. Also manually enter a course schedule, an assessment, a deadline, and a score using the interface's formats. Inspect field labels and date-format instructions. Save and reopen each item; edit the coursework name and due date, then reopen it. Create a second coursework record, then delete the first. Repeat context entry with game and rewards disabled.

Expected: Required controls are labeled and the accepted date format is stated. Saved items reproduce entered and edited values as planning context, including with game and rewards disabled. The deleted record is absent; the other records are unchanged.

C2 Manual-entry Validation (F21–23, I14)

Fixture & Approach: Test a blank course, blank assignment name, missing due date, incorrectly formatted date, whitespace-only name, and impossible calendar date independently and together. Repeat while editing an existing valid record.

Expected: The system leaves invalid records unsaved and identifies each invalid field with an explanation of the validation rules. The system rejects edits and preserves the previously saved valid record. Correcting all invalid fields allows the system to save the record. Confirm whitespace handling and calendar date validation in the interface specification before execution.

C3 Valid CSV Import and Interface Guidance (F24–25, F54, I6–8)

Fixture & Approach: Prepare CSV fixtures for course schedules, assessments, deadlines, and scores using all required columns and formats. Include at least 3 distinct valid records per file and a name with a correctly quoted comma. Inspect import instructions, import each file, and compare saved planning-context values with the source. Repeat with game and rewards disabled.

Expected: The interface identifies all required columns and each accepted format before import. Saved records match the source values. The displayed accepted count equals the fixture's record count and the rejected count is zero, including with game and rewards disabled.

C4 Mixed Valid/Invalid CSV Records (F25–27, F30, I8–9)

Fixture & Approach: Prepare a CSV with at least 5 records (including at least 2 invalid records, at least 1 missing a required value, at least 1 with an invalid format, and at least 1 containing multiple validation errors). Import the CSV, then inspect the displayed results and the stored records.

Expected: The system imports the valid records and leaves the invalid ones unsaved. The displayed accepted and rejected counts match the valid and invalid fixture totals. The interface identifies each rejected record by its location in the file and the reasons for the listed rejections.

C5 Invalid CSV Files and Duplicate Records (F28–29)

Fixture & Approach: Import a file missing a required column and a malformed file that the CSV parser is unable to parse, in separate runs. Import a valid file containing multiple records identical in every required field, and at least one record that differs from the others in one required field. Attempt to import a header-only file and an empty file.

Expected: Files with missing columns and unparseable files do not add records nor alter existing data. The duplicate pair creates one coursework record, and the interface imports the distinct valid record. Files do not create records without coursework data. Record how empty files are handled and how duplicate records affect counts before assessing the interface details.

C6 Saved-state Restoration (F31–32, P1–3)

Fixture & Approach: Save coursework, plans A and B, scheduled tasks, mixed completion states, and known game/reward values. Change and save at least one value in each category. End the application session, then begin a new one using the same test DB. Test a fresh DB with no saved records.

Expected: The new application session restores the most recently saved categorical values. Older values do not replace the latest saved values, and there are no duplicate records. Displayed progress matches the saved completion states. A fresh DB opens without phantom records or restoration errors.

C7 Task Relationships and Combined Schedule (F33–35, F40–41, I10)

Fixture & Approach: Retain different identifiers and plan associations while providing A1 and B1 the same visible task name. Display them in a combined schedule. Reschedule A1 to another date/time, then delete A2 from plan A. Restart the application, then inspect the schedule and stored associations.

Expected: Distinct records represent both plans. Each record displays its associated course or plan name. Rescheduling changes only A1 and preserves its association with plan A. Deleting A2 removes it from the combined schedule. B1 remains unchanged. The consequent associations and schedule persist after restarting the application.

C8 Overlap Detection (F36–38, I11)

Fixture & Approach: Keep A1 scheduled on one date from 10:00–11:00. Create or reschedule B1 on that date to each interval below (in separate runs):

1. 09:00–10:00 (ends when A1 starts)
2. 09:30–10:30 (partial overlap)
3. 09:30–11:30 (contains A1)
4. 10:00–11:00 (identical interval)
5. 10:15–10:45 (contained within A1)
6. 10:30–11:30 (partial overlap from the other direction)
7. 11:00–12:00 (starts when A1 ends)

Test 10:00–11:00 on a different date. Run overlapping cases within one plan, then across different plans.

Expected: Intervals 2, 3, 4, 5, and 6 trigger conflict notifications and identify both tasks involved with a conflict indicator on each. The system does not report conflict between the ‘touching’ and different date intervals. Results follow the F36 requirement overlap rule.

C9 Multiple Conflicts and Resolution (F37–39, I11)

Fixture & Approach: Schedule A1 from 10:00–11:00 and B1 from 11:00–12:00 on the same date. Create or reschedule A2 to 10:30–11:30, and resolve the conflict by moving A2 to 12:00–13:00.

Expected: The initial operation indicates the conflict between A1, A2, and B1 and displays the indicators. The interface allows the student user to resolve the conflicts through rescheduling. The schedule shows the new time and no remaining conflicts among tasks after the resolution.

C10 Completion and Plan-Specific Progress (F42, F44–45, P3)

Fixture & Approach: Start with A1 and A2 as incomplete, and B1 as complete. Complete A1, repeat the completion action, complete A2, then mark A1 incomplete. Inspect the displayed progress in the plans and the saved task states after the operations and after restarting the application.

Expected: Only plan A tasks affect plan A progress, and the completed task count changes from 0 to 1, remains one on the repeated action, becomes 2, then returns to 1. Progress from plan B remains unchanged. Incomplete tasks do not contribute to completion progress.

C11 Reward Correctness (F31–32, F43, F56)

Fixture & Approach: Enable the game and give A1 an associated reward. Record rewards and earned moves, then complete A1. Repeat the action, refresh, and restart. Deliver the same completion event twice where automated tests are available. Mark A1 incomplete and complete it again, then restart once more.

Expected: The first completion grants the associated reward and one earned game move. Repeated actions, duplicate events, refreshes, and restarts grant neither again. Marking A1 incomplete and completing it again also adds no reward or move. The saved reward totals and processed task identifier persist across restarts.

C12 Academic-data Isolation and Optional Rewards (F46, F48)

Fixture & Approach: Save a snapshot of coursework, plans, and scheduled tasks. Change each implemented data type for the game and rewards through a controlled test fixture. Compare the coursework with the snapshot, disable the game and reward features, and then compare all academic records again. Continue editing, scheduling, and completing tasks with those features disabled.

Expected: Changing game or reward data does not modify coursework records. Disabling the features does not delete or alter coursework, plans, or scheduled tasks. Planning remains available.

C13 Response Time Requirements (P4–8)

Fixture & Approach: Use the saved plans and tasks in C12, along with a CSV containing 500 distinct valid coursework records. Record the test computer, application version, browser, DB size, and timing method. Use browser automation or timestamped instrumentation to measure from the user's submitted action to the visible result. Repeat each operation at least 3 times while retaining the measurements.

Expected:

The following operations should execute within 2 seconds:

1. Requesting a stored plan displays it.
2. Creating, modifying, or deleting a scheduled task updates the displayed scheduling.
3. Adding or rescheduling a conflict task displays all conflicts.
4. Saving changes to coursework, a plan, or a scheduled task displays a confirmation (reopening verifies that the system saved the changes).

Importing the CSV with 500 records completes the import and validation and accepts all records within 5 seconds.

Game and reward cases

S1 Game setup, turns, and legal moves (F55, F57–F60)

Fixture & Approach: Start with an empty board and use a controlled random-choice fixture to exercise each possible starter. Restore independent fixtures for each check. On the student's turn, provide one earned move and play an empty cell; inspect the computer reply. Repeat a student turn with zero earned moves. Attempt moves into an occupied cell, outside the board, and during the computer's turn.

Expected: Folio provides one-player tic-tac-toe against the computer on a 3-by-3 board. New-game starter selection uses the random choice and supports either player. A legal student move consumes one earned move and places one mark. With no earned move, the student forfeits the turn and the computer moves. Invalid moves leave the board unchanged. The computer makes a legal move on its turn unless the game has ended.

S2 Results, cumulative totals, and automatic restart (F61–F63)

Fixture & Approach: Use legal near-finished boards for student and computer wins in each row, column, and diagonal, plus a full-board tie with no winning line. Record existing win, loss, and tie totals before finishing each game. Inspect the result and automatically created next game, then refresh and restart the application.

Expected: Each fixture reports the correct win, loss, or tie and increments only its matching cumulative total once. A full board with a winning line reports the winner rather than a tie. A finished game accepts no further moves; a new game starts automatically on a cleared board. Prior result totals persist without recounting the finished game.

S3 Rewards separate from game moves (F64)

Fixture & Approach: Record the configured XP, point, level, badge, and cosmetic eligibility rules. Use independent reward fixtures just below each applicable threshold and complete eligible, previously unrewarded tasks. Record rewards and earned moves before and after, then spend one earned move in the game.

Expected: XP, points, levels, badges, and cosmetic rewards can each be earned according to their configured rules. These rewards remain separate from earned game moves: spending a move does not spend or remove them, and XP or points do not substitute for a move.